Organic Cume, January 21, 2005

1. Give the energy values for the HMO wavefunctions of the following compound and calculate the overall energy of cyclobutadiene in Hückel terms.

cyclobutadiene

2. Show with molecular orbital coefficients, how cyclobutadiene can be Möbius aromatic.

3. Calculate the HMO' stabilization energy for naphthalene. The data given below should help you.

The HMO energy levels for naphthalene are:
α-2.303β; α-1.618β; α-1.303β; α-1.000β; α-0.618β; α+0.618β; α+1.000β; α+1.303β; α+1.618β; α+2.303β.

Energy Values for Reference Bond Types			
Bond type	Hess-Schaad Value (β)	Bond type	Moyano-Paniagua value (β)
H_2C=CH	2.000	H_2C=CH-CH=	2.2234
HC=CH	2.070	H_2C=CH-C=	2.2336
H_2C=C	2.000	=CH-CH=CH-CH=	2.5394
HC=C	2.108	=CH-CH=CH-C=	2.5244
C=C	2.172	=C-CH=CH-C=	2.4998
HC-CH	0.466	H_2C=C-	2.4320
HC-C	0.436	-CH=CH-	2.7524
C-C	0.436	-C=C-	2.9970

(table from Carey and Sundberg, page 532)